Level 3

Lists, Tuples, and Dictionaries

Indexing, Iteration, Dictionary Lookups

Question1: Create a list of 10 hacker tools: ["Nmap", "Metasploit", "Wireshark", "Burp Suite","JohnTheRipper", ...]

```
hacker_tools = [
    "Nmap",
    "Sqlmap",
    "Metasploit",
    "OSINT",
    "Wireshark",
    "Burp Suite",
    "JohnTheRipper",
    "OpenVPN",
    "Virustotal",
    "Hashcat"
]
```

Question2: Print the 3rd item in the list.

```
print(hacker_tools[2])
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 1.py
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 2.py
Metasploit
```

Question3: Create a dictionary of HTTP status codes (e.g., {200: "OK", 404: "Not Found"}).

```
status_codes = {
    200: "OK",
    201: "Created",
    404: "Not Found",
    500: "Internal Server Error",
}
```

Question4: Count how many times each letter appears in a string.

```
input_string = input("Enter a string: ").lower()
letter_count = {}
for char in input_string:
    if char.isalpha():
        if char in letter_count:
            letter_count[char] += 1
        else:
            letter_count[char] = 1
print(letter_count)
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 4.py
Enter a string: Nada
{'n': 1, 'a': 2, 'd': 1}
```

Question5: Sort a list of random numbers without using .sort().

```
list_of_numbers = [5, 2, 9, 1, 5, 6]
sorted_list = []

while list_of_numbers:
    min_value = min(list_of_numbers)
    sorted_list.append(min_value)
    list_of_numbers.remove(min_value)
print(sorted_list)
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 5.py
[1, 2, 5, 5, 6, 9]
```

Question6: Store ports and services in a dictionary and allow user queries.

```
ports_and_services = {
    22: "SSH",
    80: "HTTP",
    443: "HTTPS",
    21: "FTP",
    25: "SMTP"
}

input_port = int(input("Enter a port number to look up the
service: "))
if input_port in ports_and_services:
    print(f"Port {input_port} is used for
{ports_and_services[input_port]}.")
else:
    print(f"Port {input_port} is not in the dictionary.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 6.py
Enter a port number to look up the service: 22
Port 22 is used for SSH.
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 6.py
Enter a port number to look up the service: 33
Port 33 is not in the dictionary.
```

Question7: Function to remove duplicates from a list.

```
list_with_duplicates = [1, 2, 2, 3, 4, 4, 5]

def remove_duplicates(input_list):
    unique_items = []
    for item in input_list:
        if item not in unique_items:
            unique_items.append(item)
    return unique_items

print(remove_duplicates(list_with_duplicates))
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 7.py
[1, 2, 3, 4, 5]
```

Question8: Convert a list into a comma-separated string.

```
items = ["apple", "banana", "cherry"]
comma_separated_string = ", ".join(str(item) for item in
items)
print(comma_separated_string)
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 8.py
apple, banana, cherry
```

Question9: Find the longest word in a list.

```
words = ["apple", "banana", "watermelon"]
longest_word = max(words, key=len)
print(f"The longest word is: {longest_word}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 9.py
The longest word is: watermelon
```

Question10: Given a dictionary of usernames and passwords, print the password for a given username.

```
credentials = {
    "alice": "password123",
    "bob": "securepass",
    "eve": "phising123"
}

input_username = input("Enter a username: ")
if input_username in credentials:
    print(f"The password for {input_username} is:
{credentials[input_username]}")
else:
    print(f"Username {input_username} not found.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level3 % python3 10.py
Enter a username: bob
The password for bob is: securepass
```